

Social Quantum Field Theory 20~30 Volume III



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Chapter 20

Quantization of Social Fields

20.1 Introduction

The preceding chapters established the mathematical foundation of Social Quantum Field Theory (SQFT) by introducing relational state spaces, structural Hamiltonians, open-system dynamics, renormalization, information geometry, and topological organization. Throughout Volume I, the social field was treated primarily as a continuous dynamical entity whose evolution could be described through density operators and effective interactions.

A natural question now arises:

If a social field is regarded as a mathematical field, what constitutes its elementary excitation?

In quantum field theory, particles are not fundamental objects. Rather, they are localized excitations of underlying quantum fields. Electrons are excitations of the electron field, photons are excitations of the electromagnetic field, and phonons are collective excitations of crystalline lattices.

SQFT adopts an analogous—but purely mathematical—perspective.

Individuals are not interpreted as isolated units existing independently of the relational field. Instead, they are modeled as localized excitations of an evolving relational field whose interactions collectively generate observable social structures.

This reinterpretation does **not** imply that people are quantum particles. It instead provides a mathematically consistent language in which local agency and global structure coexist within a single formalism.

20.2 The Relational Vacuum

Definition 20.1 (Relational Vacuum)

A **relational vacuum** is the reference configuration of a relational field in which no localized excitations are present.

Denote this state by

$$|0\rangle. \quad (20.1)$$

The relational vacuum does **not** represent an empty society.

Instead, it represents the lowest-energy configuration of the mathematical model,

$$H|0\rangle = E_0|0\rangle, \quad (20.2)$$

where

- institutional constraints remain encoded,
- interaction rules still exist,
- potential relationships remain available,
- but no explicit local excitation has yet emerged.

Thus the vacuum contains structural potential rather than empirical emptiness.

20.3 Local Excitations

Definition 20.2 (Social Excitation)

Let

$$a_i^\dagger$$

denote the creation operator associated with relational mode i .

A localized excitation is defined as

$$|i\rangle = a_i^\dagger |0\rangle. \quad (20.3)$$

In SQFT,

$$|i\rangle$$

should be interpreted as

the mathematical representation of a locally activated relational state.

Depending upon the empirical application, such an excitation may correspond to

- an individual,
- an institution,
- an organization,
- a technological platform,
- a communication hub,
- or any localized relational entity.

20.4 Multi-Agent Configurations

Multiple localized excitations generate collective relational configurations,

$$|\Psi\rangle = a_1^\dagger a_2^\dagger \cdots a_N^\dagger |0\rangle. \quad (20.4)$$

Unlike classical set theory,

the ordering and interaction among excitations influence the resulting field configuration.

The global state therefore cannot generally be reconstructed by considering each excitation independently.

Definition 20.3 (Occupation Number)

For each relational mode,

$$N_i = a_i^\dagger a_i. \quad (20.5)$$

The expectation value

$$\langle N_i \rangle \quad (20.6)$$

measures the intensity of activation associated with relational mode i .

In empirical applications, this quantity may correspond to measures such as participation intensity, communication activity, institutional engagement, or other observable indicators, depending on the context.

20.5 Commutation Structures

The algebra satisfied by creation and annihilation operators determines the statistical properties of the relational field.

For bosonic-type collective excitations,

$$[a_i, a_j^\dagger] = \delta_{ij}, \quad (20.7)$$

where

$$[A, B] = AB - BA.$$

For fermionic-type exclusion dynamics,

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \quad (20.8)$$

where

$$\{A, B\} = AB + BA.$$

Remark

These algebraic choices should **not** be interpreted as implying that people obey Bose–Einstein or Fermi–Dirac statistics. Rather, they provide mathematical templates for modeling different classes of relational constraints, such as unrestricted participation versus exclusive occupancy.

20.6 Field Expansion

Let

$$\Phi(x, t)$$

denote the relational field.

Its operator expansion is

$$\Phi(x, t) = \sum_k \left(a_k u_k(x, t) + a_k^\dagger u_k^*(x, t) \right), \quad (20.9)$$

where

- $u_k(x, t)$ are relational mode functions,
- $a_k^\dagger$ creates localized activations,
- a_k removes localized activations.

The basis functions u_k may represent geographical regions, institutional sectors, communication channels, or other domain-specific relational modes.

20.7 Structural Hamiltonian

Using the quantized field representation, the structural Hamiltonian becomes

$$H = \sum_k \omega_k a_k^\dagger a_k + \frac{1}{2} \sum_{ijkl} V_{ijkl} a_i^\dagger a_j^\dagger a_k a_l. \quad (20.10)$$

The first term represents intrinsic activation costs, while the second encodes nonlinear interactions among relational modes.

Proposition 20.1

If

$$V_{ijkl} = 0,$$

then every relational excitation evolves independently.

Proof

With the interaction tensor absent, the Hamiltonian reduces to a sum of commuting occupation operators. Consequently, each mode evolves according to its own intrinsic frequency ω_k , and no coupling exists between distinct modes.

20.8 Interpretation

The quantization of the social field provides a mathematically precise description of how localized entities emerge from an underlying relational structure. It replaces the conceptual dichotomy between "individual" and "society" with a unified framework in which local excitations and global fields are complementary aspects of the same mathematical object.

This formulation prepares the way for the next chapter, where interactions among localized excitations will be analyzed using perturbative methods and relational interaction vertices, extending SQFT toward a fully dynamical theory of collective processes.

Chapter 21

Interaction Vertices and Perturbation Theory of Social Fields

21.1 Introduction

The previous chapter introduced the quantization of relational fields by representing localized entities as excitations of an underlying mathematical field. While this formulation explains the existence of individual relational modes, it does not yet describe how these modes interact.

In physical quantum field theory, interactions are encoded by **interaction vertices**, where elementary excitations exchange energy and momentum. In Social Quantum Field Theory (SQFT), interaction vertices are reinterpreted as mathematical representations of relational events through which information, influence, resources, or institutional constraints are transmitted.

The objective of this chapter is to construct a perturbative framework capable of describing complex relational interactions while preserving the operator formalism established in previous chapters.

21.2 Free and Interaction Hamiltonians

The structural Hamiltonian is decomposed into

$$H = H_0 + H_{\text{int}}, \quad (21.1)$$

where

- H_0 describes independent relational evolution,
- H_{int} describes interactions among relational excitations.

For the free dynamics,

$$H_0 = \sum_i \omega_i a_i^\dagger a_i, \quad (21.2)$$

while the interaction term is written as

$$H_{\text{int}} = \sum_{ijkl} g_{ijkl} a_i^\dagger a_j^\dagger a_k a_l, \quad (21.3)$$

where

$$g_{ijkl}$$

is the **relational coupling tensor**, measuring the strength of interactions among relational modes.

Definition 21.1 (Relational Coupling Constant)

A relational coupling constant quantifies the degree to which one relational excitation influences another through the underlying field.

Unlike physical coupling constants, its empirical interpretation depends upon the application and may represent

- communication efficiency,
 - institutional influence,
 - economic dependence,
 - political negotiation,
 - technological interoperability,
 - cultural transmission.
-

21.3 Perturbative Expansion

Exact solutions of

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle \quad (21.4)$$

are generally unavailable.

Assume

$$\|H_{\text{int}}\| \ll \|H_0\|. \quad (21.5)$$

The evolution operator may then be expanded as

$$U(t) = e^{-iHt/\hbar} = U_0(t) + U_1(t) + U_2(t) + \dots, \quad (21.6)$$

where

$$U_0(t) = e^{-iH_0t/\hbar} \quad (21.7)$$

represents the free evolution.

Higher-order terms successively incorporate increasingly complex relational interactions.

21.4 Dyson Series

Introducing the interaction picture,

$$H_I(t) = U_0^\dagger H_{\text{int}} U_0, \quad (21.8)$$

the evolution operator becomes

$$U_I(t) = T \exp \left[-\frac{i}{\hbar} \int H_I(t') dt' \right], \quad (21.9)$$

where

T

denotes time ordering.

Expanding the exponential,

$$\begin{aligned} U_I(t) = & 1 - \frac{i}{\hbar} \int H_I dt \\ & + \left(-\frac{i}{\hbar} \right)^2 \frac{1}{2} \int H_I H_I dt_1 dt_2 + \dots \end{aligned} \quad (21.10)$$

Each successive order corresponds to increasingly indirect relational pathways.

21.5 Interaction Vertices

Every interaction tensor

$$g_{ijkl}$$

defines a relational vertex.

Graphically,



should be interpreted as a mathematical interaction event connecting four relational modes.

Examples include

- negotiations among organizations,
- financial transactions,
- scientific collaborations,
- information diffusion,
- institutional coordination.

The diagram is a bookkeeping device rather than a literal depiction of social processes.

Definition 21.2 (Relational Vertex)

A relational vertex is an elementary interaction through which localized excitations exchange information or influence under the rules specified by the interaction Hamiltonian.

21.6 Scattering Interpretation

Suppose two relational excitations,

$$|A, B\rangle,$$

interact.

After structural evolution,

$$|A, B\rangle \rightarrow |C, D\rangle. \quad (21.11)$$

The transition amplitude is

$$A = \langle C, D|U|A, B \rangle. \quad (21.12)$$

Its squared magnitude,

$$P = |A|^2, \quad (21.13)$$

gives the probability of observing the corresponding relational outcome within the model.

Examples include

- successful policy agreements,
- formation of strategic alliances,
- dissolution of partnerships,
- emergence of collaborative communities.

21.7 Effective Interactions

Microscopic interactions often aggregate into simpler macroscopic descriptions.

The effective Hamiltonian may therefore be written as

$$H_{\text{eff}} = H_0 + H_{\text{macro}}, \quad (21.14)$$

where

$$H_{\text{macro}}$$

summarizes many unresolved microscopic interactions.

This mirrors the philosophy of effective field theory: only degrees of freedom relevant at the chosen observational scale are modeled explicitly.

21.8 Weak and Strong Coupling

Interaction strength determines the applicability of perturbation theory.

Weak Coupling

If

$$g \ll 1, \quad (21.15)$$

low-order perturbative calculations are generally sufficient.

Typical examples include

- occasional communication,
- infrequent cooperation,
- loose organizational coordination.

Strong Coupling

If

$$g \sim 1 \quad \text{or larger,} \quad (21.16)$$

higher-order interactions become important, and perturbation theory may fail.

Examples include

- tightly integrated institutions,
- highly polarized political environments,
- strongly coupled financial systems.

In such cases, non-perturbative techniques become necessary.

Theorem 21.1 (Perturbative Validity)

Assume

$$\|H_{\text{int}}\| < \|H_0\|.$$

Then the perturbative expansion converges within a neighborhood of the free theory.

Proof

The Dyson series defines a power series in the interaction Hamiltonian. Standard results from operator perturbation theory establish convergence when the interaction norm remains sufficiently small relative to the free Hamiltonian, ensuring that successive corrections decrease in magnitude.

Corollary 21.1

Systems exhibiting weak relational coupling can often be approximated accurately using only the first few perturbative orders, whereas strongly coupled systems require non-perturbative methods.

21.9 Discussion

Interaction vertices provide the mathematical mechanism by which local relational excitations generate collective dynamics. They bridge the gap between isolated entities and emergent social structures, allowing complex organizational behavior to be represented through systematically constructed interaction terms.

The perturbative framework developed in this chapter is particularly useful for analyzing systems in which interactions are relatively weak or localized. In the following chapter, this formalism will be extended beyond perturbation theory to investigate **collective phenomena arising from strongly coupled relational fields**, including spontaneous structural organization, symmetry breaking, and emergent institutional order.

Chapter 22

Spontaneous Symmetry Breaking and Emergent Institutional Order

22.1 Introduction

The perturbative framework developed in the previous chapter successfully describes weakly interacting relational systems. Many real-world social systems, however, exhibit behavior that cannot be understood as small perturbations around equilibrium.

Examples include

- the sudden emergence of political consensus,
- rapid institutional collapse,
- formation of scientific paradigms,
- spontaneous market standards,
- cultural norm formation,
- technological lock-in.

These phenomena exhibit a common characteristic:

Global order emerges without any centralized controller.

In modern physics, analogous behavior is described through **spontaneous symmetry breaking (SSB)**.

In SQFT, spontaneous symmetry breaking is reinterpreted as a mathematical description of **collective institutional ordering** arising from relational interactions.

22.2 Symmetric Relational States

Suppose the relational Hamiltonian possesses a continuous symmetry group

G .

Then

$$[H, U(g)] = 0, \quad g \in G, \quad (22.1)$$

where

$$U(g)$$

represents the symmetry transformation.

Examples include

- equivalent institutional alternatives,
- identical strategic choices,
- interchangeable communication channels.

Initially,

no alternative is mathematically preferred.

Definition 22.1 (Structural Symmetry)

A relational system possesses structural symmetry if its governing equations remain invariant under a specified class of relational transformations.

Importantly,

symmetry concerns the mathematical description,

not necessarily the observed outcome.

22.3 Degenerate Ground States

Suppose the effective potential

$$V(\phi)$$

takes the form

$$V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4, \quad \lambda > 0. \quad (22.2)$$

The minimum satisfies

$$|\phi| = \frac{\mu}{\sqrt{2\lambda}}. \quad (22.3)$$

Instead of a unique minimum,

there exists an entire family of equivalent minima,

forming

$$M_0.$$

Each minimum represents a possible institutional configuration.

Interpretation

The mathematics does **not** prescribe which equilibrium will be realized.

Rather,

historical fluctuations,

local interactions,

or external disturbances

select one particular minimum.

22.4 Institutional Ordering

Suppose an initially disordered system evolves toward one equilibrium,

$$\phi = \phi_0. \quad (22.4)$$

Although

the governing equations remain symmetric,

the observed configuration no longer exhibits the full symmetry.

This phenomenon constitutes **institutional ordering**.

Examples include

- adoption of a common language,
 - emergence of monetary standards,
 - acceptance of scientific conventions,
 - dominance of a technological platform,
 - stabilization of legal institutions.
-

Definition 22.2 (Order Parameter)

An **order parameter**

$$\Phi$$

is a measurable quantity satisfying

$$\Phi = 0$$

in the disordered phase,

and

$$\Phi \neq 0$$

after structural ordering.

$$\Phi = \langle \hat{O} \rangle. \quad (22.5)$$

Depending upon the empirical system,

the order parameter may represent

- institutional legitimacy,
- consensus strength,
- market concentration,
- technological adoption,
- organizational coherence.

22.5 Collective Modes

Near equilibrium,

small fluctuations satisfy

$$\phi = \phi_0 + \delta\phi. \quad (22.6)$$

These fluctuations correspond to

collective relational modes.

Unlike isolated agents,

collective modes describe

shared institutional dynamics,

such as

- public opinion waves,
- organizational learning,
- coordinated innovation,
- cultural diffusion.

22.6 Goldstone-Type Modes

If a continuous symmetry is spontaneously broken,

low-energy collective excitations naturally emerge.

Within SQFT,

these may be interpreted as

persistent relational fluctuations requiring little external effort to propagate.

Examples include

- established social conventions,
- linguistic evolution,
- informal organizational norms,
- cooperative habits.

These collective modes often spread efficiently because they evolve along nearly flat regions of the effective relational landscape.

22.7 Explicit Symmetry Breaking

Real societies rarely possess perfect symmetry.

External influences introduce additional terms,

$$H = H_0 + H_{\text{int}} + H_{\text{ext}}. \quad (22.7)$$

Examples include

- legislation,
- economic sanctions,
- technological innovation,
- environmental shocks,
- military intervention.

These contributions bias the selection of equilibrium,

favoring certain institutional configurations over others.

22.8 Structural Phase Diagram

As external parameters vary,

qualitatively different relational phases may emerge.

Let

g

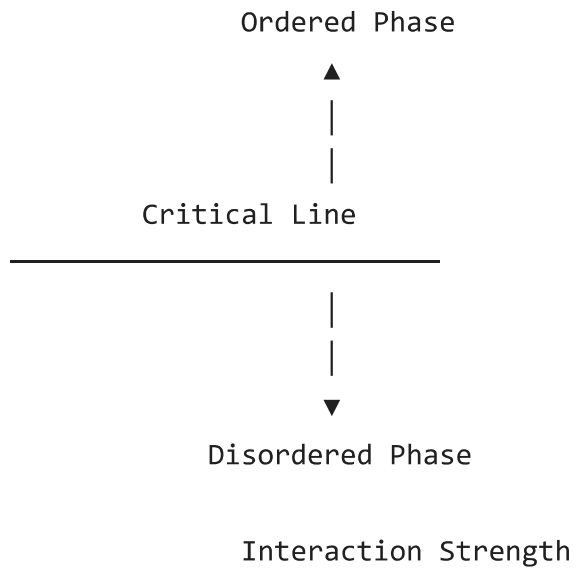
denote the interaction strength,

and

T

represent a generalized social fluctuation parameter.

The resulting phase structure may be represented schematically.



Crossing the critical boundary changes

the qualitative organization

of the relational field.

22.9 Criticality

Near the transition,

the correlation length

ξ

increases dramatically,

$$\xi \rightarrow \infty. \quad (22.8)$$

Consequently,

local disturbances may influence the entire relational field.

Examples include

- financial contagion,
 - viral information cascades,
 - nationwide political movements,
 - rapid technological diffusion.
-

Theorem 22.1 (Emergent Institutional Order)

Suppose

1. the relational Hamiltonian possesses a continuous symmetry;
2. the effective potential contains multiple degenerate minima;
3. the system evolves toward one stable minimum.

Then

the observable relational configuration exhibits spontaneous symmetry breaking, despite the symmetry of the governing equations.

Proof

The Hamiltonian remains invariant under the symmetry group GG , so all minima related by the symmetry are energetically equivalent. Dynamical evolution or fluctuations lead the system to occupy one particular minimum. Once selected, observables are evaluated with respect to that configuration rather than the entire symmetric manifold. Consequently, the realized state no longer displays the full symmetry of the governing equations, even though the equations themselves remain unchanged.

Corollary 22.1

Persistent institutions may be interpreted as stable ordered phases of the relational field rather than as fixed entities existing independently of social dynamics.

22.10 Discussion

Spontaneous symmetry breaking provides a mathematical explanation for how coherent institutional structures can arise from decentralized interactions without requiring centralized design. The emergence of laws, norms, standards, and conventions is modeled as the selection of one among many mathematically equivalent equilibria through the dynamics of the relational field.

This perspective shifts the focus from individual decisions to the stability of collective order. In the next chapter, this framework will be extended to **critical phenomena and renormalization near institutional phase transitions**, where small-scale relational interactions generate large-scale structural transformations across multiple levels of organization.

Chapter 23

Critical Phenomena and Renormalization in Social Quantum Field Theory

23.1 Introduction

The previous chapter demonstrated that spontaneous symmetry breaking provides a mechanism through which ordered institutional structures emerge from decentralized relational interactions. However, the transition between disorder and order is rarely abrupt or confined to a single scale.

Historical transformations—such as financial crises, political revolutions, technological revolutions, and paradigm shifts in science—typically unfold through interactions across multiple organizational levels. Local events accumulate, amplify, and eventually reshape global structures.

Understanding this multiscale behavior requires a mathematical framework capable of linking microscopic relational dynamics with macroscopic institutional phenomena.

In theoretical physics, this role is played by the **Renormalization Group (RG)**.

Within Social Quantum Field Theory (SQFT), renormalization is reinterpreted as a mathematical procedure that describes how relational structures evolve when observed at different organizational scales.

23.2 Scale Dependence

No empirical system appears identical when examined at different levels of resolution.

Consider the hierarchy

$$\text{Individual} \Rightarrow \text{Organization} \Rightarrow \text{Institution} \Rightarrow \text{Society}. \quad (23.1)$$

Each level possesses its own effective variables.

Consequently,

the mathematical description should depend explicitly upon the observational scale,

$$\mu.$$

Definition 23.1 (Observation Scale)

An **observation scale** is the characteristic level at which relational interactions are represented.

Typical scales include

- individual interactions,
- organizational processes,
- sector-wide coordination,
- national institutions,
- global relational systems.

23.3 Effective Theory

Suppose

$$H(\mu)$$

denotes the effective Hamiltonian observed at scale

$$\mu.$$

Then

$$H(\mu) = H_{\text{relevant}} + H_{\text{irrelevant}}. \quad (23.2)$$

Only sufficiently important degrees of freedom remain explicitly represented, while microscopic details are absorbed into effective parameters.

Interpretation

An economist studying international trade need not model every consumer.

Likewise,

an organizational theorist need not model every conversation occurring inside a corporation.

Renormalization formalizes this simplification mathematically.

23.4 Coarse-Graining

Let

$$\rho(x)$$

denote a relational field.

A coarse-grained field is obtained through averaging,

$$\rho(x) = \int K(x - y) \rho(y) dy, \quad (23.3)$$

where

is an averaging kernel.

Microscopic fluctuations are suppressed,

while large-scale structures remain.

Definition 23.2 (Coarse-Graining)

Coarse-graining is the mathematical operation that replaces detailed relational configurations with effective macroscopic descriptions.

It preserves large-scale organization while eliminating unnecessary microscopic complexity.

23.5 Running Couplings

Interaction strengths generally depend upon scale.

Let

$$g(\mu)$$

denote a scale-dependent coupling.

Its evolution satisfies

$$\mu \frac{dg}{d\mu} = \beta(g), \quad (23.4)$$

where

$$\beta(g)$$

is the **beta function**.

Interpretation

Examples include

- institutional influence varying across organizational levels,
 - communication efficiency changing with network size,
 - market competition differing between local and global economies,
 - governance mechanisms evolving across administrative scales.
-

23.6 Fixed Points

A fixed point satisfies

$$\beta(g^*) = 0. \quad (23.5)$$

At a fixed point,

the effective theory becomes invariant under changes of observational scale.

Definition 23.3 (Structural Fixed Point)

A **structural fixed point** is a relational configuration whose effective dynamics remain unchanged under coarse-graining.

Such configurations represent scale-invariant institutional organization.

Examples include

- mature constitutional systems,
 - stabilized scientific paradigms,
 - globally adopted technological standards,
 - long-lived financial infrastructures.
-

23.7 Universality

One of the most remarkable consequences of renormalization theory is **universality**.

Different microscopic systems may exhibit identical large-scale behavior.

Within SQFT,

distinct societies,

organizations,

or technological ecosystems

may converge toward common structural laws despite substantial historical differences.

Symbolically,

$$\text{Different Microscopic Dynamics} \implies \text{Same Macroscopic Universality Class.} \quad (23.6)$$

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23.8 Critical Exponents

Near a structural transition,

observable quantities often obey power laws.

For example,

$$\xi \propto |g - g_c|^{-\nu}, \quad (23.7)$$

where

- ξ is the correlation length,
- g_c is the critical coupling,
- ν is a critical exponent.

Similarly,

an order parameter satisfies

$$\Phi \propto (g - g_c)^\beta. \quad (23.8)$$

These exponents characterize the universality class of the transition rather than the details of individual systems.

23.9 Institutional Criticality

As the correlation length grows,

$$\xi \rightarrow \infty, \quad (23.9)$$

small disturbances propagate across increasingly large regions of the relational field.

Empirical manifestations include

- cascading financial failures,
- nationwide protest movements,
- rapid technological diffusion,
- widespread organizational restructuring,
- viral information propagation.

Near criticality,

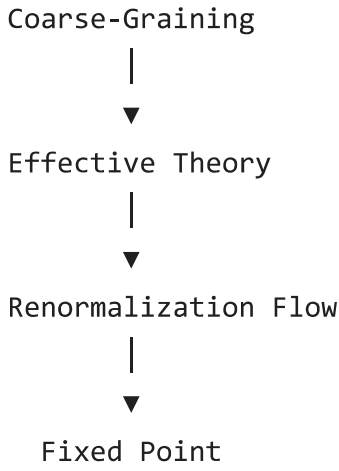
local interventions may produce disproportionately large systemic consequences.

23.10 Renormalization Flow

The evolution of effective theories may be visualized as a trajectory through parameter space.

Microscopic Theory





Different initial conditions may converge toward the same fixed point, illustrating the emergence of universality.

Theorem 23.1 (Scale Invariance at Fixed Points)

Let

$$g^*$$

satisfy

$$\beta(g^*) = 0.$$

Then the corresponding effective relational theory is invariant under infinitesimal changes in observational scale.

Proof

The beta function measures the variation of the coupling constant with respect to the logarithm of the observation scale. At a fixed point,

$$\mu \frac{dg}{d\mu} = 0,$$

so the effective coupling remains constant under scale transformations. Since the Hamiltonian depends only on these couplings, the effective dynamics are unchanged by infinitesimal coarse-graining. Therefore, the theory exhibits scale invariance at the fixed point.

Corollary 23.1

Institutional configurations associated with structural fixed points possess enhanced robustness against small-scale perturbations, although sufficiently large external disturbances may still drive the system toward a different universality class.

23.11 Discussion

Renormalization provides the mathematical bridge between local interactions and global social organization. Rather than treating institutions as static entities, SQFT views them as effective theories emerging from successive levels of coarse-graining.

This perspective explains why systems with very different historical origins can nevertheless display similar large-scale patterns of stability, crisis, and adaptation. It also establishes a principled framework for analyzing structural change across scales without requiring every microscopic interaction to be modeled explicitly.

The next chapter extends this multiscale perspective to **topological phases of relational systems**, where global organizational properties are determined not by local parameters alone but by the overall topology of the relational field.

Chapter 24

Topological Phases and Topological Protection in Relational Systems

24.1 Introduction

Previous chapters have shown that collective relational systems can exhibit emergent order through spontaneous symmetry breaking and multiscale organization through renormalization. Nevertheless, certain social structures display a remarkable form of robustness that cannot be adequately explained by local interactions alone.

Examples include

- constitutional traditions that survive repeated political transitions,
- scientific methodologies that persist across generations,
- legal systems that withstand changes of government,
- monetary standards maintained through economic crises,
- engineering protocols that remain stable despite technological evolution.

Such persistence suggests that some organizational properties depend not on local details but on the **global connectivity** of the relational field.

In modern condensed matter physics, analogous phenomena are described as **topological phases**.

Within Social Quantum Field Theory (SQFT), topology provides a mathematical framework for understanding why certain institutional structures remain stable under substantial local

perturbations.

24.2 Relational Topology

Consider a relational network represented by

$$G = (V, E), \quad (24.1)$$

where

- V denotes relational entities,
- E denotes relational interactions.

The topology of the system is determined not by the precise strengths of the interactions but by the global pattern of connectivity.

Definition 24.1 (Relational Topology)

The **relational topology** of a system is the collection of connectivity properties that remain invariant under continuous deformations preserving relational continuity.

Examples of continuous deformations include

- gradual changes in communication intensity,
- incremental organizational restructuring,
- moderate fluctuations in participation levels.

Topology changes only when the connectivity structure itself is fundamentally altered.

24.3 Topological Invariants

A topological invariant is a quantity preserved under continuous structural deformation.

Let

$$T(G)$$

denote such an invariant.

Then

$$T(G_1) = T(G_2) \quad (24.2)$$

whenever

$$G_1 \sim G_2$$

are topologically equivalent.

Possible invariants include

- connected components,
- cycles,
- homology groups,
- Betti numbers,
- persistent homology.

Definition 24.2 (Institutional Topological Invariant)

An institutional topological invariant is a global structural property that remains unchanged despite continuous modifications of local relational interactions.

24.4 Betti Numbers

Let

$$b_k$$

denote the k -th Betti number.

Examples include

$$b_0 = \text{number of connected components}, \quad (24.3)$$

$$b_1 = \text{number of independent cycles}, \quad (24.4)$$

$$b_2 = \text{number of enclosed voids}. \quad (24.5)$$

Within SQFT,

these quantities characterize global organizational architecture rather than individual relationships.

Interpretation

For example,

a high value of

$$b_1$$

may indicate

- multiple redundant communication pathways,
- overlapping institutional collaborations,

- resilient organizational feedback loops.

Such redundancy often enhances structural robustness.

24.5 Persistent Homology

Real relational systems evolve continuously.

Instead of studying a single network,

consider a filtration

$$G(\epsilon), \quad (24.6)$$

parameterized by

$$\epsilon.$$

Persistent homology tracks topological features as

$$\epsilon$$

changes.

Features persisting across a wide interval

$$[\epsilon_1, \epsilon_2]$$

are interpreted as structurally significant rather than accidental fluctuations.

Definition 24.3 (Persistent Structure)

A persistent structure is a topological feature that survives across multiple observational scales or relational thresholds.

Persistent structures often correspond to

- enduring alliances,
 - long-term institutional arrangements,
 - stable scientific communities,
 - resilient economic ecosystems.
-

24.6 Topological Protection

One of the most remarkable consequences of topology is **topological protection**.

Certain global properties cannot be destroyed by small local perturbations.

Within SQFT,

this principle suggests

$$\text{Local disturbance} \Rightarrow \text{Global institutional collapse.} \quad (24.7)$$

Instead,

a qualitative structural change requires a genuine topological transition.

Examples

Topological protection may help explain the persistence of

- constitutional governance,
- internationally accepted measurement standards,
- programming language ecosystems,
- internet communication protocols,
- scientific peer-review institutions.

These systems may adapt locally while preserving their global organizational identity.

24.7 Topological Phase Transition

A topological transition occurs when

$$T$$

changes discontinuously,

$$T_1 \neq T_2. \quad (24.8)$$

Unlike conventional symmetry-breaking transitions,

topological transitions may occur without a local order parameter.

Instead,

they involve changes in global connectivity.

Interpretation

Examples include

- fragmentation of international alliances,
 - collapse of communication infrastructures,
 - emergence of new technological standards,
 - reorganization of global supply chains.
-

24.8 Homotopy Classes

Continuous deformations define equivalence classes.

Denote the fundamental group by

$$\pi_1(G). \quad (24.9)$$

Distinct homotopy classes represent qualitatively different organizational architectures, even when local interaction statistics appear similar.

24.9 Topological Stability

Suppose

$$\delta H$$

is a sufficiently small perturbation.

Then

$$H \rightarrow H + \delta H. \quad (24.10)$$

If

$$T$$

remains unchanged,

the system preserves its topological phase despite local modifications.

Consequently,

observable behavior may remain stable even though many microscopic interactions evolve over time.

Theorem 24.1 (Topological Robustness)

Let

$$G$$

be a relational system possessing topological invariant

$$T.$$

If continuous perturbations preserve

$$T,$$

then the system remains within the same topological phase.

Proof

By definition, topological invariants are unchanged under continuous deformations that do not alter global connectivity. Since the perturbation preserves the invariant, the system remains in the same equivalence class. Therefore, although local relational parameters may vary, the global organizational phase is unchanged.

Corollary 24.1

Institutional resilience is not determined solely by the strength of individual relationships. It also depends upon the preservation of the global topological organization of the relational field.

24.10 Discussion

Topology introduces a new level of description into Social Quantum Field Theory. Earlier chapters focused on local interactions, symmetry, and scale dependence. This chapter demonstrates that some forms of institutional stability arise from properties that transcend local dynamics altogether.

By distinguishing between geometric change and topological change, SQFT explains how organizations, legal systems, technological ecosystems, and scientific communities can undergo continuous adaptation while maintaining a stable global identity.

In the next chapter, these topological ideas will be combined with **information geometry**, yielding a unified framework in which curvature, topology, and information jointly determine the evolution and stability of complex relational systems.

Chapter 25

Information Geometry and the Curvature of Relational Fields

25.1 Introduction

The previous chapter established that global organizational stability may arise from topological invariants that remain unchanged under continuous structural deformations. Topology explains *whether* a relational structure belongs to the same global phase, but it does not quantify *how* neighboring relational configurations differ or how rapidly they evolve.

To describe these questions, Social Quantum Field Theory (SQFT) introduces **information geometry**.

Information geometry equips the space of relational states with a metric, allowing notions such as

- distance,
- angle,
- curvature,
- geodesic paths,
- statistical distinguishability,

to be defined mathematically.

Within SQFT, information geometry provides a bridge between statistical inference and relational dynamics, enabling structural evolution to be interpreted as motion on a curved information manifold.

25.2 The Relational Statistical Manifold

Let

$$\theta = (\theta^1, \theta^2, \dots, \theta^n)$$

denote the parameter vector describing a relational system.

Examples include

- institutional trust,
- communication efficiency,
- collaboration density,
- innovation rate,
- organizational resilience.

The collection of all admissible parameter values forms the **relational statistical manifold**

$$M. \quad (25.1)$$

Each point on

$$M$$

corresponds to one possible macroscopic state of the relational field.

Definition 25.1 (Relational Manifold)

The **Relational Manifold** is the differentiable manifold consisting of all statistically distinguishable relational configurations.

25.3 Fisher Information Metric

Suppose

$$P(x|\Theta)$$

denotes the probability distribution describing empirical observations.

The Fisher Information Matrix is

$$g_{ij} = E \left[\frac{\partial \ln P}{\partial \theta^i} \frac{\partial \ln P}{\partial \theta^j} \right]. \quad (25.2)$$

This metric defines the infinitesimal distance

$$ds^2 = g_{ij} d\theta^i d\theta^j. \quad (25.3)$$

Large distances correspond to statistically distinguishable relational states,

whereas nearby points represent configurations that are difficult to differentiate empirically.

Interpretation

A small displacement on the manifold corresponds to

- gradual institutional reform,
- incremental technological change,
- slow cultural evolution.

Large displacements correspond to

- revolutions,
 - systemic crises,
 - rapid organizational transformation.
-

25.4 Geodesics

Given the metric

$$g_{ij},$$

the shortest path between two relational states satisfies

$$\frac{d^2 \theta^k}{ds^2} + \Gamma_{ij}^k \frac{d\theta^i}{ds} \frac{d\theta^j}{ds} = 0, \quad (25.4)$$

where

$$\Gamma_{ij}^k$$

are the Christoffel symbols.

Definition 25.2 (Optimal Structural Evolution)

A **geodesic** represents the locally shortest trajectory through relational state space.

Within SQFT,

such trajectories may model

- efficient institutional adaptation,
 - minimal-cost organizational reform,
 - optimal policy transitions.
-

25.5 Curvature

The geometry of

$$M$$

is characterized by the Riemann curvature tensor

$$R^i_{jkl}. \quad (25.5)$$

Contracting indices yields

the Ricci tensor

$$R_{ij}, \quad (25.6)$$

and the scalar curvature

$$R. \quad (25.7)$$

Interpretation

Large positive curvature often indicates

- highly constrained relational evolution,
- strong institutional rigidity,
- reduced adaptive flexibility.

Small curvature indicates

- flexible organizational dynamics,
- multiple adaptive pathways.

Negative curvature may correspond to

- rapidly diverging developmental trajectories,
- heightened sensitivity to perturbations,
- competitive diversification.

25.6 Information Distance

The distinguishability between two nearby relational configurations satisfies

$$D^2 = g_{ij} \Delta \theta^i \Delta \theta^j. \quad (25.8)$$

Unlike Euclidean distance,

information distance measures statistical separability rather than physical separation.

Example

Two governments with different constitutions may nevertheless occupy nearby positions in information space if their observable institutional behavior is statistically similar.

Conversely,

systems with nearly identical legal structures may lie far apart if their empirical dynamics differ substantially.

25.7 Entropic Dynamics

Suppose structural evolution follows the direction of maximal information gain.

Then

$$\frac{d\theta^i}{dt} = -g^{ij} \frac{\partial S}{\partial \theta^j}, \quad (25.9)$$

where

$$S$$

denotes relational entropy.

This equation describes gradient flow on the information manifold.

Definition 25.3 (Entropic Flow)

An **entropic flow** is the trajectory that maximizes the increase (or decrease, depending on the convention) of an entropy-related objective under the constraints of the information geometry.

25.8 Curvature and Structural Instability

Rapid increases in curvature often precede structural transitions.

Let

$$K = |R|. \quad (25.10)$$

If

$$K \rightarrow \infty,$$

the relational manifold develops strong geometric singularities,

indicating that neighboring trajectories diverge rapidly.

Potential empirical manifestations include

- institutional breakdown,
- financial instability,
- organizational fragmentation,
- rapid polarization.

25.9 Geometric Interpretation of Institutional Resilience

Institutional resilience may be interpreted geometrically.

Systems possessing

- moderate curvature,
- multiple geodesic pathways,
- stable topological invariants,

can adapt to disturbances while preserving overall structural integrity.

Conversely,

systems with

- extremely high curvature,
- few admissible trajectories,
- topological fragility,

are more susceptible to abrupt structural transitions.

Theorem 25.1 (Geodesic Stability)

Suppose

M

is a complete Riemannian manifold equipped with Fisher metric

$$g_{ij}.$$

If the sectional curvature remains uniformly bounded,

$$|K| < K_{\max}, \quad (25.11)$$

then sufficiently nearby relational trajectories remain continuously dependent upon their initial conditions over finite evolution intervals.

Proof

The bounded-curvature assumption ensures that geodesic deviation remains finite over finite parameter intervals. Standard results from Riemannian geometry imply continuous dependence of geodesic solutions on initial data. Consequently, small perturbations of relational states produce proportionally small deviations in subsequent evolution, establishing local structural stability.

Corollary 25.1

Relational systems evolving on smoothly curved information manifolds exhibit greater predictability than systems whose geometric structure contains regions of extreme curvature or singular behavior.

25.10 Discussion

Information geometry complements the topological perspective developed in the previous chapter. Whereas topology identifies global structural invariants, geometry quantifies the local shape of the relational landscape and the cost of moving between alternative configurations.

Together,

- **operator theory** describes dynamics,
- **renormalization** describes scale dependence,
- **topology** describes global organization,
- **information geometry** describes local structure.

These four mathematical components form the core analytical architecture of Social Quantum Field Theory.

The next chapter will integrate these ideas with **quantum information concepts**, introducing relational entropy, mutual information, and correlation measures as quantitative indicators

of collective organization and information flow within complex relational fields.

Chapter 26

Quantum Information Theory and Relational Information Dynamics

26.1 Introduction

Previous chapters established that relational systems possess

- dynamical evolution,
- multiscale organization,
- topological stability,
- information geometry.

These mathematical structures describe **how** relational systems evolve.

The remaining question is

How is information stored, distributed, and transmitted throughout a relational field?

Classical information theory treats information as localized messages transmitted between independent agents.

Social Quantum Field Theory (SQFT) adopts a different mathematical viewpoint.

Information is regarded as a **property of the relational field itself**, rather than merely of individual entities.

This chapter introduces the information-theoretic foundation of SQFT by adapting concepts from quantum information theory into a purely relational mathematical framework.

26.2 Relational Density Operator

The complete relational state is represented by the density operator

$$\rho. \quad (26.1)$$

It satisfies

$$\begin{aligned} \rho &\geq 0, \\ \text{Tr}(\rho) &= 1. \end{aligned} \quad (26.2)$$

Unlike a state vector,

the density operator naturally represents

- uncertainty,
- incomplete observations,
- heterogeneous populations,
- mixed institutional configurations.

Definition 26.1 (Relational Mixed State)

A **relational mixed state** is a statistical ensemble of possible relational configurations represented by a density operator.

Such states arise whenever the observer possesses incomplete information about the underlying relational field.

26.3 Relational Entropy

The uncertainty associated with

$$\rho$$

is measured by the von Neumann entropy,

$$S(\rho) = -\text{Tr}(\rho \ln \rho). \quad (26.3)$$

Interpretation

Large entropy corresponds to

- fragmented coordination,
- institutional uncertainty,
- diversified organizational behavior,
- heterogeneous expectations.

Small entropy corresponds to

- coherent institutions,
- coordinated collective behavior,
- highly ordered relational structures.

Entropy therefore measures informational organization rather than social "disorder" in a normative sense.

26.4 Reduced Relational States

Suppose the total system consists of two interacting subsystems,

$$A \text{ and } B.$$

A and B.

The reduced density operator for subsystem

$$A$$

is

$$\rho_A = \text{Tr}_B(\rho). \quad (26.4)$$

Similarly,

$$\rho_B = \text{Tr}_A(\rho). \quad (26.5)$$

Partial tracing removes degrees of freedom belonging to the complementary subsystem while preserving the observable statistics of the subsystem under consideration.

Interpretation

Examples include

- one government within an international system,
- one company within a market,
- one university within a scientific collaboration network.

Each subsystem remains mathematically influenced by interactions with the larger relational field.

26.5 Mutual Information

The total shared information between

$$A$$

and

$$B$$

is

$$I(A:B) = S(A) + S(B) - S(AB). \quad (26.6)$$

Mutual information is always non-negative,

$$I(A:B) \geq 0. \quad (26.7)$$

Definition 26.2 (Relational Mutual Information)

Relational mutual information quantifies the statistical dependence between two relational subsystems.

Unlike direct communication,

it measures shared structure,

whether or not explicit interaction is immediately observable.

Examples

Large mutual information may indicate

- coordinated policy development,
 - synchronized financial markets,
 - scientific collaboration,
 - technological interoperability,
 - cultural similarity.
-

26.6 Relative Entropy

Suppose two relational configurations satisfy

$$\rho \text{ and } \sigma.$$

and

Their distinguishability is measured by

$$D(\rho||\sigma) = \text{Tr}(\rho \ln \rho - \rho \ln \sigma). \quad (26.8)$$

Interpretation

Relative entropy measures

how different one institutional configuration is from another.

Applications include

- comparing organizational reforms,
 - evaluating policy alternatives,
 - measuring structural divergence,
 - detecting systemic change.
-

26.7 Relational Coherence

Beyond entropy,

SQFT introduces the concept of **relational coherence**.

Let

$$\rho = \rho_{\text{diag}} + \rho_{\text{off}}.$$

Define

$$C = \sum_{i \neq j} |\rho_{ij}|. \quad (26.9)$$

Large coherence indicates

- synchronized expectations,
- coordinated institutions,
- coherent organizational behavior.

Low coherence indicates

- fragmented communication,
- decentralized adaptation,
- institutional incoherence.

Unlike quantum coherence in microscopic physics, relational coherence is interpreted purely as a mathematical indicator of collective coordination.

26.8 Information Flow

Information changes over time according to

$$\frac{dS}{dt}. \quad (26.10)$$

Positive values indicate increasing uncertainty,

while negative values indicate increasing coordination.

More generally,

information currents may satisfy

$$\nabla_\mu J^\mu = \Sigma, \quad (26.11)$$

where

- J^μ denotes the information current,
 - Σ represents local information production or dissipation.
-

26.9 Information Conservation

Closed relational systems satisfy

$$\frac{dS_{\text{total}}}{dt} = 0. \quad (26.12)$$

Open systems satisfy

$$\frac{dS_{\text{total}}}{dt} \neq 0, \quad (26.13)$$

because external environments continually exchange information with the relational field.

This naturally connects with the Lindblad dynamics introduced in earlier chapters.

Theorem 26.1 (Non-Negativity of Relational Mutual Information)

For any bipartite relational state,

$$I(A:B) \geq 0.$$

Proof

The von Neumann entropy satisfies the property of subadditivity,

$$S(AB) \leq S(A) + S(B).$$

Rearranging the inequality immediately yields

$$S(A) + S(B) - S(AB) \geq 0.$$

Therefore,

$$I(A:B) \geq 0.$$

■

Corollary 26.1

If

$$I(A:B) = 0,$$

then the relational subsystems are statistically independent within the mathematical model.

A positive value indicates the presence of shared relational structure, though it does not by itself establish causality.

26.10 Discussion

Information theory provides the quantitative language through which relational organization can be measured. Entropy captures uncertainty, mutual information quantifies shared structure, relative entropy compares institutional configurations, and relational coherence measures the degree of coordinated organization.

These information-theoretic quantities complement the geometric and topological tools developed in previous chapters. Together they describe not only the shape and connectivity of relational fields but also the information encoded within them.

In the next chapter, these concepts will be combined into a unified framework of **Relational Entanglement Networks**, where long-range statistical dependencies among multiple subsystems are represented through graph-theoretic and tensor-network methods, extending SQFT toward large-scale computational modeling of complex relational systems.

Chapter 27

Relational Entanglement Networks and Tensor Network Representations

27.1 Introduction

The previous chapter established that relational information can be quantified through entropy, mutual information, coherence, and density operators. These quantities describe pairwise or subsystem relationships, but large-scale relational systems involve thousands or even millions of interacting entities.

Modern organizations, financial systems, scientific collaborations, and communication platforms exhibit highly structured patterns of dependency extending far beyond simple pairwise interactions.

To represent such systems efficiently, Social Quantum Field Theory (SQFT) introduces **Relational Entanglement Networks**, together with tensor-network representations that provide scalable mathematical descriptions of high-dimensional relational fields.

Throughout this chapter, the term *entanglement* refers exclusively to **non-factorizable statistical dependence within the mathematical model** and carries no implication of microscopic quantum entanglement.

27.2 Factorizable and Non-Factorizable Relational States

Consider two relational subsystems,

$$A \quad \text{and} \quad B.$$

A completely independent relational state satisfies

$$\rho = \rho_A \otimes \rho_B. \quad (27.1)$$

In contrast,

a correlated relational state satisfies

$$\rho \neq \rho_A \otimes \rho_B. \quad (27.2)$$

Definition 27.1 (Relational Entanglement)

A **relationally entangled state** is a relational state whose statistical structure cannot be decomposed into a simple tensor product of independent subsystem states.

This definition concerns mathematical dependence rather than physical quantum phenomena.

Interpretation

Relational entanglement naturally arises in

- tightly coupled supply chains,
- multinational alliances,
- scientific research communities,
- financial institutions,
- integrated technological ecosystems.

Changes affecting one subsystem alter the statistical description of others because the relational field is jointly structured.

27.3 Multipartite Relational Systems

Real relational systems contain many interacting components.

The total Hilbert space becomes

$$H = \bigotimes_{i=1}^N H_i. \quad (27.3)$$

The corresponding density operator

$$\rho$$

generally possesses high-order correlations extending across multiple organizational levels.

Direct computation rapidly becomes infeasible because

$$\dim(H) = \prod_{i=1}^N d_i, \quad (27.4)$$

which grows exponentially with system size.

27.4 Tensor Decomposition

To overcome exponential complexity,

the relational state is decomposed into tensors.

Let

$$T^{i_1 i_2 \dots i_n}$$

represent a high-order relational tensor.

Instead of storing every coefficient explicitly,

the tensor is factorized into lower-order tensors connected by internal indices.

This dramatically reduces computational complexity while preserving the dominant correlation structure.

Definition 27.2 (Tensor Network)

A **tensor network** is a collection of low-rank tensors connected through contracted indices representing relational dependencies.

27.5 Matrix Product States

For approximately one-dimensional relational structures,

the global state may be written as

$$|\Psi\rangle = \sum A^{[1]} A^{[2]} \dots A^{[N]} |i_1 i_2 \dots i_N\rangle. \quad (27.5)$$

This representation is known as a **Matrix Product State (MPS)**.

Applications

Examples include

- hierarchical organizations,
- supply chains,
- command structures,
- production pipelines,

- sequential decision processes.

Because long-range correlations are represented efficiently,

MPS provides an attractive computational framework for many organizational systems.

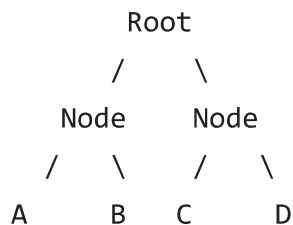
27.6 Tree Tensor Networks

Hierarchical institutions often possess branching structures.

Such systems are naturally represented by

Tree Tensor Networks (TTN),

whose graph resembles



Examples include

- governments,
- multinational corporations,
- university systems,
- military organizations.

Tree structures efficiently represent multilevel decision hierarchies.

27.7 Projected Entangled Pair States

Higher-dimensional relational systems require richer connectivity.

Projected Entangled Pair States (PEPS)

generalize MPS to networks possessing two-dimensional or higher-dimensional interactions.

Applications include

- urban communication systems,
- transportation infrastructure,
- international trade networks,
- scientific collaboration graphs.

27.8 MERA

Some relational systems evolve simultaneously across multiple organizational scales.

The **Multi-scale Entanglement Renormalization Ansatz (MERA)** combines

- tensor decomposition,
- renormalization,
- hierarchical coarse-graining.

Its layered structure naturally mirrors

- local interactions,
- organizational aggregation,
- institutional emergence,
- societal organization.

27.9 Entanglement Entropy

Suppose subsystem

A

is embedded within a larger relational field.

Its entanglement entropy is

$$S_A = -\text{Tr}(\rho_A \ln \rho_A). \quad (27.6)$$

Large values indicate

strong statistical dependence upon the surrounding relational environment.

Small values indicate

relative organizational independence.

Interpretation

Examples include

- multinational corporations strongly embedded within global markets,
- universities deeply integrated into international research communities,
- governments highly dependent upon global financial institutions.

27.10 Graph Representation

Tensor networks admit an equivalent graph-theoretic interpretation.

Let

$$G = (V, E).$$

Vertices represent tensors,

while edges represent contracted relational indices.

Weighted edges correspond to stronger relational correlations.

This dual interpretation connects

- tensor methods,
- graph theory,
- network science,

within a unified mathematical language.

27.11 Computational Complexity

Suppose

$$D$$

denotes the bond dimension.

Many tensor-network algorithms scale approximately as

$$O(ND^3), \quad (27.7)$$

rather than exponentially.

This reduction makes previously intractable relational simulations computationally feasible.

Theorem 27.1 (Efficient Representation of Weakly Correlated Relational Fields)

Suppose the relational correlations satisfy an area-law scaling,

$$S_A \propto |\partial A|. \quad (27.8)$$

Then there exists a tensor-network approximation whose computational complexity grows polynomially with system size.

Proof

Area-law scaling implies that relational information crossing subsystem boundaries remains bounded by the size of the boundary rather than the subsystem volume. Standard tensor-network approximation results then guarantee that low bond-dimension representations capture the dominant correlations with controllable error. Consequently, polynomial computational complexity is sufficient for accurate approximation.

Corollary 27.1

Large relational systems exhibiting localized or hierarchical correlations may be simulated efficiently without representing the full exponentially large relational state.

27.12 Discussion

Relational Entanglement Networks extend SQFT beyond pairwise interactions by providing a scalable mathematical framework for representing high-dimensional dependence across complex systems. Tensor-network methods unify ideas from operator theory, renormalization, topology, and information theory into a computationally tractable architecture capable of modeling organizations, markets, technological ecosystems, and scientific communities.

In the next chapter, this computational framework will be integrated with **Bayesian inference and statistical learning**, allowing empirical data to estimate Hamiltonians, coupling tensors, and latent relational states, thereby connecting the theoretical formalism of SQFT with quantitative model calibration and prediction.

Chapter 28

Bayesian Inference and Statistical Learning in Social Quantum Field Theory

28.1 Introduction

The preceding chapters established the mathematical foundations of Social Quantum Field Theory (SQFT), including operator dynamics, information geometry, topology, renormalization, and tensor-network representations. These developments define a theoretical framework capable of describing complex relational systems.

A scientific theory, however, becomes practically useful only when its mathematical parameters can be estimated from empirical observations.

This chapter introduces **Bayesian inference** as the statistical bridge between SQFT and observational data. Rather than treating model parameters as fixed but unknown constants, Bayesian methods regard them as probability distributions that are continually updated as new evidence becomes available.

Within SQFT, Bayesian inference provides a mathematically rigorous procedure for estimating structural Hamiltonians, interaction tensors, latent relational states, and dynamical parameters directly from empirical measurements.

28.2 Unknown Relational Parameters

Suppose the relational Hamiltonian depends upon an unknown parameter vector

$$\Theta = (\theta_1, \theta_2, \dots, \theta_n). \quad (28.1)$$

Typical parameters include

- interaction strengths,
- institutional coupling coefficients,
- diffusion rates,
- dissipation constants,
- external driving fields.

Rather than assuming

$$\Theta$$

is exactly known,

SQFT models uncertainty explicitly.

Definition 28.1 (Parameter Distribution)

A **parameter distribution** is a probability density over admissible values of

$$\Theta,$$

representing the observer's uncertainty before or after empirical observations.

28.3 Prior Knowledge

Before observing new data,

the parameter distribution is

$$P(\Theta). \quad (28.2)$$

This **prior distribution** incorporates

- previous empirical studies,
- theoretical constraints,
- historical observations,
- expert knowledge.

The prior may be

- Gaussian,
- uniform,
- hierarchical,
- sparsity-promoting,

depending upon the application.

28.4 Likelihood Function

Suppose observational data

D

are collected.

Given parameters

θ ,

their probability is

$$P(D|\theta). \quad (28.3)$$

This quantity is called the **likelihood**.

Within SQFT,

the likelihood may be constructed from

- network observations,
- institutional statistics,
- temporal trajectories,
- communication records,
- market indicators.

28.5 Bayes' Theorem

Bayesian updating is governed by

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}. \quad (28.4)$$

where

$$P(\theta|D)$$

is the posterior distribution.

Interpretation

Posterior distributions represent revised knowledge after incorporating empirical evidence.

Repeated observations continually refine parameter estimates.

28.6 Maximum A Posteriori Estimation

One practical estimator is

$$\hat{\theta}_{MAP} = \operatorname{argmax}_{\theta} P(\theta|D). \quad (28.5)$$

Unlike maximum likelihood estimation,

MAP incorporates prior knowledge,

making it particularly useful when data are limited.

28.7 Bayesian Updating of Hamiltonians

Suppose the structural Hamiltonian is

$$H(\theta).$$

$H(\theta)$.

After new observations,

its parameter distribution becomes

$$P(H|D). \quad (28.6)$$

Consequently,

the Hamiltonian itself becomes progressively refined,

allowing SQFT to evolve with accumulating empirical evidence.

Definition 28.2 (Adaptive Hamiltonian)

An **adaptive Hamiltonian** is a structural Hamiltonian whose parameters are continually updated through Bayesian inference as additional observations become available.

28.8 Sequential Learning

Suppose observations arrive sequentially,

$$D_1, D_2, \dots, D_n.$$

Bayesian updating proceeds recursively,

$$P(\Theta|D_1, \dots, D_n) \propto P(D_n|\Theta)P(\Theta|D_1, \dots, D_{n-1}). \quad (28.7)$$

This recursive formulation enables

- online learning,
- streaming data analysis,
- real-time forecasting.

28.9 Bayesian Networks and Relational Fields

Probabilistic graphical models naturally complement SQFT.

A Bayesian network

$$G = (V, E)$$

represents conditional dependencies among relational variables.

Unlike tensor networks,

which encode high-dimensional correlations,

Bayesian networks explicitly represent probabilistic causation under specified model assumptions.

Together,

the two approaches provide complementary views of relational structure.

28.10 Uncertainty Quantification

Every parameter estimate possesses uncertainty.

Bayesian inference naturally provides

credible intervals,

posterior variances,

and predictive distributions.

Rather than reporting

$$\theta = 2.1,$$

one reports

$$\theta = 2.1 \pm 0.3. \quad (28.8)$$

This explicit treatment of uncertainty is particularly valuable when analyzing complex relational systems.

28.11 Bayesian Prediction

Suppose future observations satisfy

$$D_{\text{future}}.$$

D_{future} .

The predictive distribution is

$$P(D_{\text{future}}|D) = \int P(D_{\text{future}}|\theta)P(\theta|D) d\theta. \quad (28.9)$$

This equation averages predictions over all plausible parameter values, thereby incorporating parameter uncertainty into forecasting.

Theorem 28.1 (Posterior Consistency)

Suppose

1. the SQFT model is correctly specified;
2. observations are independent or satisfy appropriate mixing conditions;
3. the prior assigns positive probability to the true parameter neighborhood.

Then,

as the amount of observational data increases,

the posterior distribution concentrates around the true parameter values.

Proof

Under standard Bayesian asymptotic theory, posterior consistency follows from the identifiability of the statistical model together with regularity conditions ensuring convergence of the likelihood. Since the prior places positive probability on neighborhoods of the true parameter, increasing data progressively overwhelm prior uncertainty, causing the posterior distribution to converge toward the true parameter values.

Corollary 28.1

As empirical evidence accumulates, uncertainty in the estimated Hamiltonian, coupling tensors, and dynamical parameters decreases, improving the predictive accuracy of the SQFT model provided that the underlying assumptions remain valid.

28.12 Discussion

Bayesian inference transforms SQFT from a purely theoretical framework into an empirically calibratable scientific model. It provides systematic procedures for estimating unknown parameters, quantifying uncertainty, and updating theoretical descriptions in light of new evidence.

Combined with tensor-network computation, renormalization methods, and information geometry, Bayesian learning enables SQFT to support data-driven analyses of large-scale relational systems while preserving mathematical rigor.

In the next chapter, these statistical methods will be integrated with **machine learning and physics-informed artificial intelligence**, establishing a computational framework in which theoretical constraints and observational data jointly guide the discovery and prediction of complex relational dynamics.

Chapter 29

Physics-Informed Artificial Intelligence for Social Quantum Field Theory

29.1 Introduction

The previous chapter demonstrated how Bayesian inference enables the parameters of Social Quantum Field Theory (SQFT) to be estimated from empirical observations.

Bayesian methods provide statistical consistency, uncertainty quantification, and principled model updating. Nevertheless, modern relational systems often generate data on a scale that exceeds the capabilities of classical statistical inference alone.

Examples include

- global financial transactions,
- social communication platforms,
- scientific publication networks,
- multinational supply chains,
- large-scale governmental databases.

These systems produce relational observations containing billions of interacting variables.

Artificial Intelligence (AI), particularly modern machine learning, provides powerful computational tools for discovering patterns within such data. However, purely data-driven learning frequently lacks interpretability and may violate established theoretical principles.

SQFT therefore adopts a different philosophy:

Artificial intelligence should not replace mathematical theory; it should operate within it.

Accordingly, this chapter develops a **Physics-Informed Artificial Intelligence (PIAI)** framework in which machine learning algorithms are constrained by the mathematical structure of Social Quantum Field Theory.

29.2 Data-Driven Versus Theory-Driven Learning

Traditional machine learning seeks a function

$$f_{\theta}(x), \quad (29.1)$$

that minimizes prediction error,

$$L = \sum_i \left(y_i - f_{\theta}(x_i) \right)^2. \quad (29.2)$$

Although highly successful,

such models often

- ignore conservation laws,
- violate structural constraints,
- produce physically or mathematically inconsistent predictions.

Definition 29.1 (Theory-Constrained Learning)

A **Theory-Constrained Learning System** is a learning algorithm whose optimization is restricted by the governing equations of SQFT.

Instead of minimizing only empirical error,

the optimization also satisfies structural equations derived from the theory.

29.3 Physics-Informed Loss Functions

Suppose observational data produce the empirical loss

$$L_{\text{data}}.$$

The SQFT equations introduce an additional structural loss

$$L_{\text{SQFT}}.$$

The complete optimization objective becomes

$$L = L_{\text{data}} + \lambda L_{\text{SQFT}}, \quad (29.3)$$

where

$$\lambda$$

balances empirical accuracy against theoretical consistency.

Examples of Structural Constraints

The theoretical loss may enforce

- Lindblad evolution,

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + D(\rho),$$

- normalization,

$$\text{Tr}(\rho) = 1,$$

- positivity,

$$\rho \geq 0,$$

- conservation laws,
- symmetry constraints,
- topological invariants.

29.4 Neural Approximation of Relational Fields

Let

$$\Phi(x, t)$$

denote the relational field.

A neural network approximates

$$\Phi_{\theta}(x, t). \quad (29.4)$$

The parameters

$$\theta$$

are optimized simultaneously using

- observational data,
- SQFT evolution equations,
- information-theoretic constraints.

Unlike unconstrained neural networks,

the resulting approximation respects the underlying mathematical structure.

29.5 Graph Neural Networks

Most relational systems naturally possess graph structure.

Let

$$G = (V, E).$$

Each node contains feature vector

$$h_i.$$

Graph Neural Networks (GNNs) update node representations via

$$h_i^{(l+1)} = f\left(h_i^{(l)}, \{h_j^{(l)} : j \in N(i)\}\right). \quad (29.5)$$

Within SQFT,

the neighborhood aggregation function is constrained by the interaction Hamiltonian,

rather than being learned entirely from data.

Interpretation

Graph Neural Networks become computational approximations of relational field dynamics rather than purely statistical predictors.

29.6 Hamiltonian Neural Networks

Instead of learning arbitrary dynamics,

Hamiltonian Neural Networks approximate

$$H(\Theta). \quad (29.6)$$

The evolution equations become

$$\frac{dq}{dt} = \frac{\partial H}{\partial p},$$

$dt dq = \partial p \partial H,$

$$\frac{dp}{dt} = -\frac{\partial H}{\partial q}. \quad (29.7)$$

Within SQFT,

these variables correspond to generalized relational coordinates and conjugate momenta.

Learning the Hamiltonian directly preserves structural consistency throughout prediction.

29.7 Operator Learning

Rather than approximating trajectories,

modern neural operators learn mappings

$$F: H \rightarrow \rho(t). \quad (29.8)$$

Such operator-learning methods

approximate entire solution operators,

allowing efficient simulation across many initial conditions.

29.8 Reinforcement Learning

Institutional decision-making often involves sequential optimization.

Let

$$(s, a, r)$$

denote

- state,
- action,
- reward.

The objective becomes

$$\max_{\pi} E\left[\sum_t \gamma^t r_t\right]. \quad (29.9)$$

Within SQFT,

the environment evolves according to relational field equations,

ensuring that learned policies remain compatible with structural dynamics.

29.9 Explainable Artificial Intelligence

Scientific prediction requires interpretability.

Accordingly,

SQFT emphasizes

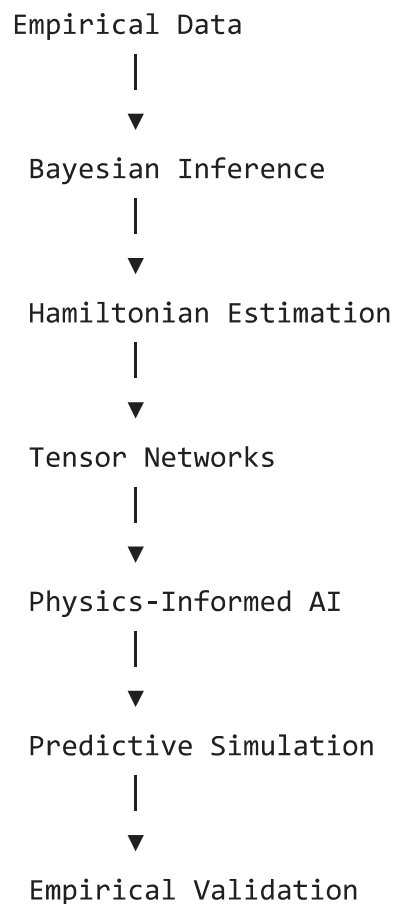
- interpretable Hamiltonians,
- identifiable coupling tensors,
- observable order parameters,
- explicit uncertainty estimates,

rather than opaque prediction alone.

Explainability ensures that AI contributes to scientific understanding rather than merely numerical accuracy.

29.10 Hybrid Computational Architecture

A complete SQFT computational platform may integrate



Each component contributes complementary capabilities,

forming a closed scientific workflow.

Theorem 29.1 (Theory-Constrained Generalization)

Suppose

1. observational data are generated by an SQFT-consistent relational process;
2. the learning model satisfies the structural constraints of SQFT;
3. optimization converges to a feasible solution.

Then the learned model possesses lower structural inconsistency than an unconstrained data-driven predictor trained on the same observations.

Proof

Because the feasible hypothesis space is restricted to models satisfying the governing equations and invariants of SQFT, optimization cannot converge to solutions violating those structural principles. Consequently, although empirical prediction error depends upon the quality of the data and optimization procedure, the resulting model exhibits reduced structural inconsistency relative to unconstrained learning algorithms. This theorem concerns theoretical consistency rather than universal superiority in predictive accuracy.

Corollary 29.1

Embedding theoretical constraints within machine learning improves the interpretability and structural validity of learned relational models, while empirical performance remains subject to the availability and quality of observational data.

29.11 Discussion

Physics-Informed Artificial Intelligence represents the computational realization of Social Quantum Field Theory. Rather than treating theory and data as competing approaches, SQFT combines them into a unified learning framework in which mathematical structure guides statistical inference and artificial intelligence accelerates empirical discovery.

This integration transforms SQFT into a continuously learning scientific system: Bayesian inference estimates parameters, tensor networks enable scalable computation, and physics-informed AI refines predictions while preserving theoretical coherence.

In the next chapter, these computational tools will culminate in a comprehensive framework for **Digital Twin Social Fields**, where real-world relational systems are represented by continuously updated mathematical twins capable of simulation, scenario analysis, and predictive decision support.

Chapter 30

Digital Twin Social Fields: Real-Time Computational Representation of Complex Relational Systems

30.1 Introduction

The preceding chapters developed the mathematical and computational foundations necessary for Social Quantum Field Theory (SQFT):

- operator dynamics,
- open-system evolution,
- renormalization,
- topology,
- information geometry,
- tensor networks,
- Bayesian inference,
- physics-informed artificial intelligence.

Together these components make possible an important new capability:

the construction of Digital Twin Social Fields.

A Digital Twin Social Field is not merely a database or simulation model.

It is a continuously updated mathematical representation of a real relational system whose internal state evolves according to the governing equations of SQFT while assimilating empirical observations in real time.

30.2 Concept of a Digital Twin

In engineering,

a digital twin represents a virtual counterpart of a physical system.

Within SQFT,

the object being replicated is not a machine,

but a relational field.

Definition 30.1 (Digital Twin Social Field)

A **Digital Twin Social Field (DTSF)** is a computational realization

$$D(t) = (\rho, H, \Theta, O) \quad (30.1)$$

whose state evolves simultaneously through

- theoretical dynamics,
- empirical observations,
- statistical updating,
- computational prediction.

Unlike conventional simulations,

the twin evolves continuously as new information becomes available.

30.3 Mathematical Architecture

The complete digital twin consists of four coupled layers.

Layer I

Empirical observations

$$D(t).$$

Layer II

Bayesian parameter estimation

$$P(\Theta|D).$$

Layer III

SQFT evolution

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + D(\rho). \quad (30.2)$$

Layer IV

Predictive simulation

$$\rho(t + \Delta t).$$

These layers operate recursively,

forming a continuously updated computational system.

30.4 State Assimilation

Suppose observations become available at time

$$t_n.$$

The predicted state is

$$\rho^-(t_n).$$

After assimilation,

the updated state becomes

$$\rho^+(t_n). \quad (30.3)$$

This process resembles sequential data assimilation used in numerical weather prediction, but is formulated here for relational field dynamics.

Definition 30.2 (Relational Assimilation)

Relational assimilation is the mathematical process through which observational data modify the estimated state of the relational field while preserving theoretical consistency.

30.5 Feedback Loop

The digital twin evolves through a closed feedback cycle,

$$\begin{aligned} & \text{Observation} \\ \Rightarrow & \text{Parameter Update} \\ \Rightarrow & \text{Field Evolution} \\ \Rightarrow & \text{Prediction} \\ \Rightarrow & \text{New Observation.} \end{aligned} \quad (30.4)$$

Each iteration improves the accuracy of the twin.

30.6 Forecasting

Suppose

$$\rho(t)$$

is known.

Future evolution satisfies

$$\rho(t + \tau) = U(\tau)\rho(t)U^\dagger(\tau). \quad (30.5)$$

For open systems,

the Lindblad evolution replaces purely unitary propagation.

Forecasts may therefore include

- institutional evolution,
- communication dynamics,
- organizational adaptation,
- systemic resilience.

30.7 Scenario Simulation

Digital twins permit counterfactual experiments.

Suppose an external intervention modifies

$$H \rightarrow H + \delta H. \quad (30.6)$$

The resulting evolution becomes

$$\rho' = U' \rho U'^{\dagger}. \quad (30.7)$$

Different interventions may then be compared quantitatively before implementation.

Examples

Possible applications include

- organizational restructuring,
- policy evaluation,
- infrastructure planning,
- technological adoption,
- supply-chain redesign.

The digital twin provides a controlled computational environment for evaluating alternative scenarios.

30.8 Early-Warning Indicators

Because the digital twin continuously estimates

- entropy,
- curvature,
- order parameters,
- coupling strengths,
- correlation lengths,

it can identify approaching structural transitions.

Possible warning indicators include

$$\frac{dS}{dt}, \quad (30.8)$$

$$|R|, \quad (30.9)$$

$$\xi, \quad (30.10)$$

and

$$\Phi. \quad (30.11)$$

Rapid changes in these quantities may indicate increasing systemic instability.

Importantly, such indicators should be validated empirically within each application domain before being relied upon for decision-making.

30.9 Uncertainty Propagation

Forecast uncertainty is represented by the posterior distribution

$$P(\rho(t)|D). \quad (30.12)$$

Future uncertainty satisfies

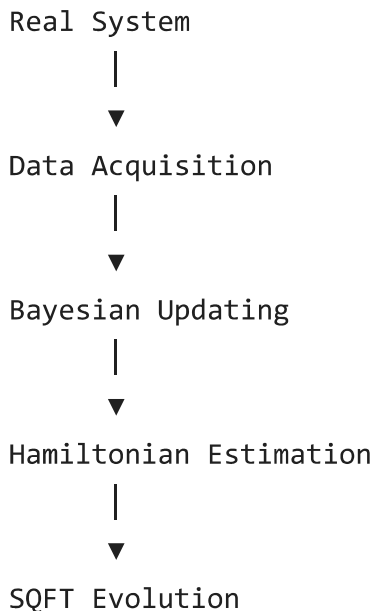
$$P(\rho(t + \tau)|D). \quad (30.13)$$

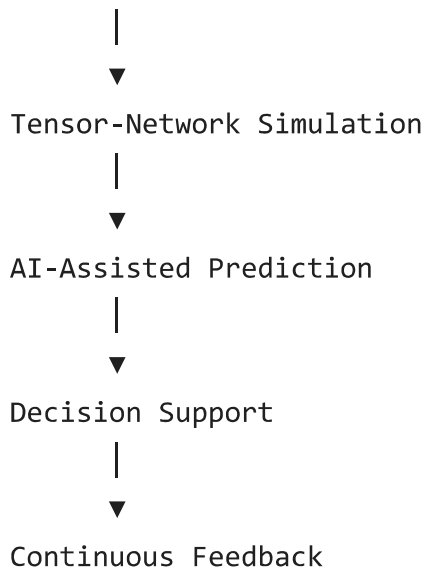
Rather than producing a single deterministic prediction,

the digital twin generates probabilistic forecasts with associated confidence estimates.

30.10 Computational Pipeline

A complete implementation may proceed as follows.





This architecture integrates all computational components introduced throughout Volume II.

Theorem 30.1 (Consistency of Sequential Digital Twin Updating)

Suppose

1. observations arrive sequentially;
2. Bayesian updating is correctly implemented;
3. the SQFT evolution operator preserves positivity and normalization of the relational state;
4. observational uncertainties are properly incorporated into the posterior distribution.

Then the Digital Twin Social Field generates a sequence of relational state estimates that remain mathematically consistent with both the governing evolution equations and the accumulated observational evidence.

Proof

Sequential Bayesian updating guarantees that each posterior distribution incorporates all previous observations together with the newest evidence. The evolution operator advances the state according to the governing SQFT dynamics while preserving admissible state properties. Alternating prediction and assimilation therefore yields a recursively consistent sequence of state estimates. This theorem establishes internal consistency of the updating procedure; predictive accuracy remains dependent upon model adequacy and data quality.

Corollary 30.1

As additional observations become available, the Digital Twin Social Field can progressively refine both its internal state estimates and its forecasts, while explicitly representing the uncertainty associated with each prediction.

30.11 Discussion

The Digital Twin Social Field represents the convergence of the theoretical and computational developments presented throughout this volume. It is not merely a visualization tool, but a mathematically grounded framework that integrates relational dynamics, statistical inference, artificial intelligence, and real-time data assimilation into a unified computational system.

Within SQFT, the digital twin serves as the operational interface between theory and practice. It enables continuous model refinement, scenario analysis, probabilistic forecasting, and hypothesis testing while remaining anchored to the mathematical principles established in earlier chapters.

The following chapter will extend this framework to **Decision Theory and Optimal Control of Relational Fields**, investigating how interventions can be designed to guide complex relational systems toward desired outcomes under uncertainty.



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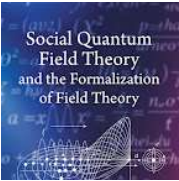
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社會量子場論（SQFT）超越布迪厄的場域理論，建構關係社會學的形式化 1~10

7月 15, 2026



預印本 社會量子場論與場論的形式化 周易先 獨立研究者 · 版本 2.0（數學版） · 2026年7月 摘要 本文提出社會量子場論（SQFT）作為用以表示複雜社會系統中關係動態的形式數學框架。SQFT並不主張社 ...

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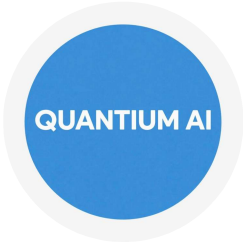
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